

MQSS Interactive Demo

11.03.2026 | Burak Mete

Creating a token for MQSS Jobs



lrz

Status

Tokens

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portal.quantum.lrz.de

14:31

Available System Resources

AQT20

AQT

20-qubit ion trap system from AQT. Seated at LRZ's QIC.

Status: Online

Qubits: 14

Quantum Technology: Ion-Trap

EQE1

BETA

IQM

54-qubit superconducting system from IQM. Seated at LRZ's Compute Cube.

Status: Online

Qubits: 53

Quantum Technology: Superconducting

MAQCS

planqc

12-qubit neutral atom demonstrator. Seated at planqc.

Status: Online

Qubits: 16

Quantum Technology: Simulator

Resources under maintenance

MUNIQ-Atoms20

20-qubit neutral atom demonstrator from Max Planck Institute of Quantum Optics (MPQ). This resource is under maintenance.

Status: Offline

Qubits: 20

Quantum Technology: Neutral Atoms

Q20

IQM

20-qubit superconducting system from IQM. Seated at LRZ's QIC.

Status: Offline

Qubits: 20

Quantum Technology: Superconducting

Q5

IQM

5-qubit superconducting system from IQM. Seated at LRZ's QIC.

Status: Offline

Qubits: 5

Quantum Technology: Superconducting

QExa20

IQM

20-qubit superconducting system from IQM. Seated at LRZ's Compute Cube.

Status: Offline

Qubits: 20

Quantum Technology: Superconducting

<https://portal.quantum.lrz.de>

Euro-Q-Exa Kick-off | Burak Mete | 11.03.2026

Using the MQSS Adapters



After creating the token:

1. Clone the repository

```
git clone git@github.com:Munich-Quantum-Software-Stack/Training-Material.git
```

2. Rename the file **.env.example** to **.env**
3. Place your token inside the field MQSS_TOKEN within **.env**
4. Install uv for dependency management (or alternative dependency managers)

```
pip install uv
```

5. Create and activate a virtual environment

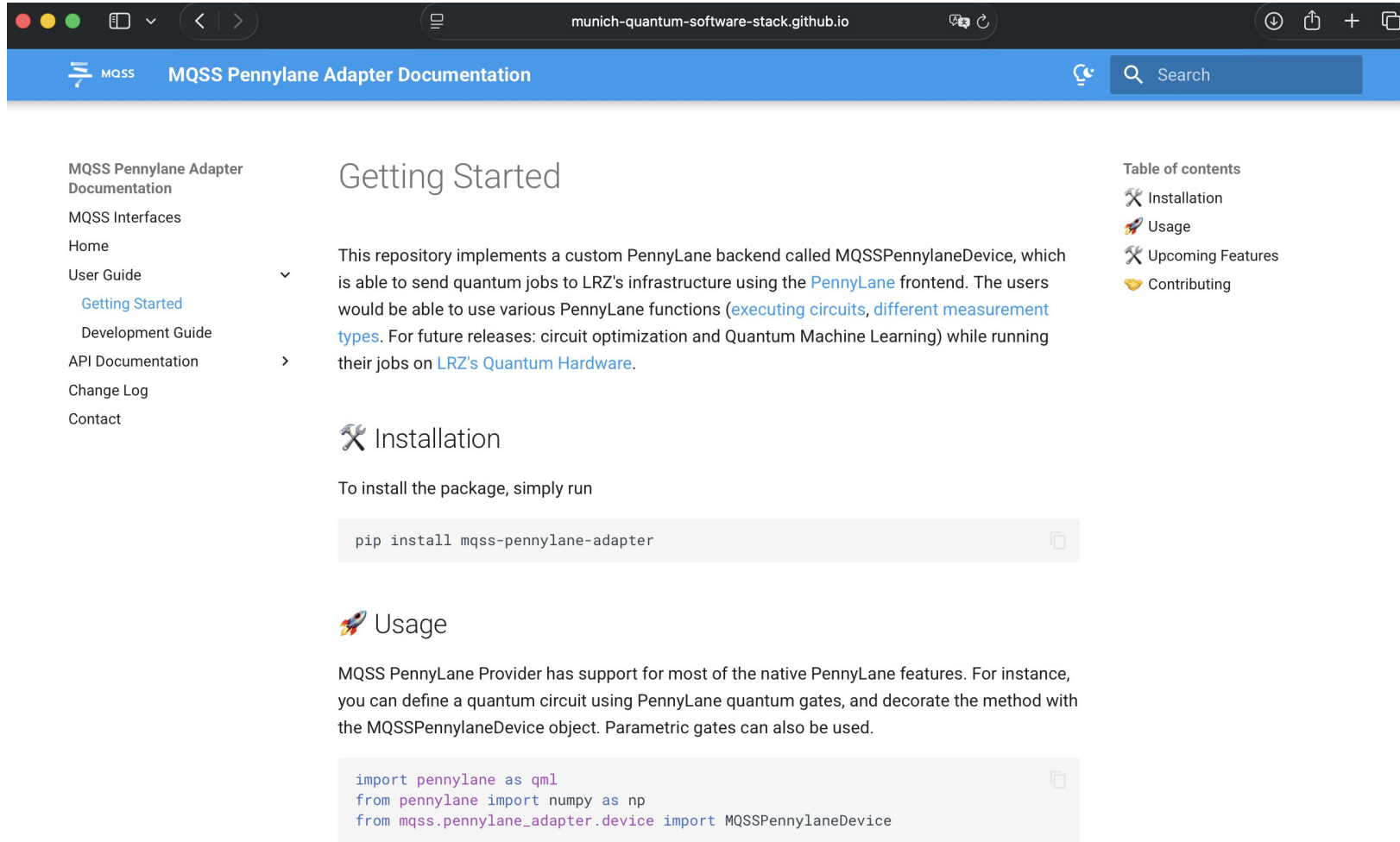
```
uv sync  
source .venv/bin/activate
```

6. Start running the examples

```
python examples/01_simple_circuit_submission.py
```



Repository for Training Materials



The screenshot shows a web browser displaying the documentation for the MQSS PennyLane Adapter. The browser's address bar shows the URL `munich-quantum-software-stack.github.io`. The page has a blue header with the title "MQSS PennyLane Adapter Documentation" and a search bar. On the left, a sidebar lists navigation links: "MQSS PennyLane Adapter Documentation", "MQSS Interfaces", "Home", "User Guide" (with a dropdown arrow), "Getting Started" (highlighted in blue), "Development Guide", "API Documentation", "Change Log", and "Contact". The main content area is titled "Getting Started" and contains a paragraph explaining that the repository implements a custom PennyLane backend called `MQSSPennyLaneDevice`, which sends quantum jobs to LRZ's infrastructure using the PennyLane frontend. It mentions that users can use various PennyLane functions (like `executing circuits` and `different measurement types`) while running jobs on LRZ's Quantum Hardware. Below this, there is an "Installation" section with a wrench icon, stating that to install the package, one should simply run the command `pip install mqss-pennylane-adapter`. Further down is a "Usage" section with a rocket icon, explaining that the MQSS PennyLane Provider supports most native PennyLane features. It gives an example of defining a quantum circuit using PennyLane quantum gates and decorating the method with the `MQSSPennyLaneDevice` object. A code block shows the following imports:

```
import pennylane as qml
from pennylane import numpy as np
from mqss.pennylane_adapter.device import MQSSPennyLaneDevice
```

. On the right side of the main content, a "Table of contents" section lists: "Installation", "Usage" (highlighted with a rocket icon), "Upcoming Features", and "Contributing".



Documentation for MQSS Adapters

MQSS Benchmarking Framework

Access and Usage (BETA)



Munich-Quantum-Software-Stack / MQSS-Benchmarking-Framework

Code Issues 9 Pull requests 1 Agents Discussions Actions Projects Security Insights Settings

MQSS-Benchmarking-Framework Public

develop 6 Branches 0 Tags

Go to file Add file Code

mdepasca-lrz More generic contact template (#26) 7dad737 · 2 weeks ago 23 Commits		
.github	More generic contact template (#26)	2 weeks ago
examples	Introduce benchmark storage system (with initial FileStor...	2 months ago
src/mqssbench	Introduce benchmark storage system (with initial FileStor...	2 months ago
test	Introduce benchmark storage system (with initial FileStor...	2 months ago
.gitignore	Introduce benchmark storage system (with initial FileStor...	2 months ago
.pre-commit-config.yaml	Initial commit (#1)	7 months ago
.python-version	Feat/bm/structural changes (#6)	5 months ago
LICENSE	Initial commit	7 months ago
README.md	Introduce benchmark storage system (with initial FileStor...	2 months ago
pyproject.toml	Introduce benchmark storage system (with initial FileStor...	2 months ago
pytest.ini	Major refactor: introduce new extensible benchmarking fr...	2 months ago
uv.lock	Introduce benchmark storage system (with initial FileStor...	2 months ago

About

Automated and reproducible benchmarking framework for quantum computing workflows.

benchmarking hpc quantum-computing hybrid-quantum-classical-algorithms hybrid-quantum-classical

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MQSS Benchmarking Framework

<https://github.com/Munich-Quantum-Software-Stack/MQSS-Benchmarking-Framework>